NAUSICAA v2

The charge-sensitive amplifier: noise, filtering and architecture

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Abstract

This note analyses the charge-sensitive amplifier for the NAUSICAA ion readout. It derives the transfer function of the amplifier, the three sources of electronic noise and their expression as an equivalent charge at the pad, the effect of integrating over a frame and of subtracting consecutive frames, and the charge bookkeeping of a two-capacitor architecture that performs that subtraction in the analogue domain. It then analyses a cascode amplifier with source degeneration, the bandwidth and stability constraints of a real amplifier, and the role of the pad capacitance. The last section compares the reset architecture with the continuous-discharge amplifier simulated and calibrated in this repository.

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1 Notation and scope

The analysis uses one set of symbols throughout.

- C_D: capacitance of the pad and its connection to the amplifier input, in farad, between 1 pF and 5 pF for the TopMetal pad. Where the amplifier's own input capacitance matters it is included in C_D.
- C_f: feedback capacitance. The present NAUSICAA circuit uses 3 fF.
- A: open-loop voltage gain of the amplifier, dimensionless and negative in sign convention, written $-A$ with $A > 0$.
- $e_n^2(f)$: one-sided spectral density of the voltage noise referred to the amplifier input, in V^2/Hz. It has a white part e_t^2 (thermal) and a flicker part A_f/f. The corner frequency $f_c = A_f/e_t^2$ is where the two are equal.
- $i_R^2 = 4kT_\mathrm{K}/R_f$: spectral density of the current noise of a feedback resistor R_f at temperature T_K, in A^2/Hz; k is Boltzmann's constant and $4kT_\mathrm{K} = 1.66 \times 10^{-20}$ J at 300 K.
- T: length of the integration window, one readout frame. The NAUSICAA productions use 20 ms, 30 ms and 40 ms.
- T_CDS: interval between the two samples that are subtracted. For consecutive frames $T_\mathrm{CDS} = T$.
- ENC: equivalent noise charge, the root-mean-square charge at the pad that would produce the observed output noise, expressed in electrons ($e = 1.602 \times 10^{-19}$ C).

Each section states a circuit or an operation, derives its result, and says what it means for NAUSICAA. Key results are boxed. The standard references for this material are Spieler [1], Radeka [2] and Gatti and Manfredi [3]. Correlated double sampling comes from CCD readout [4].

2 The charge-sensitive amplifier

The ions arriving at the pad induce a current $i(t)$. Figure 1 shows the circuit. Write v_i for the voltage at the input node and $v_o = -Av_i$ for the output. Charge conservation implies that the current i delivered by the pad is shared by the two capacitors:

$$i(t) = i_{C_D}(t) + i_{C_f}(t). \tag{1}$$

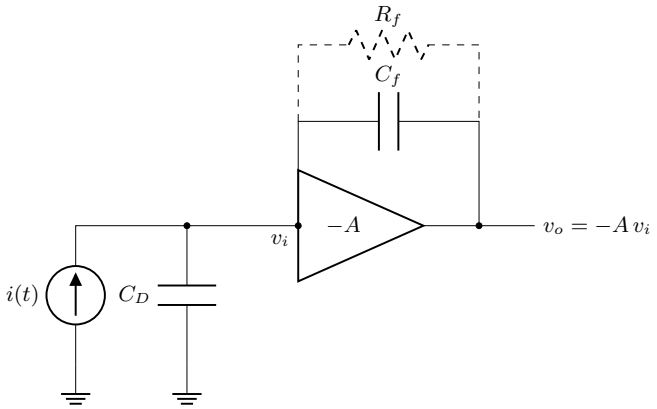


Figure 1: The pad is a current source $i(t)$ in parallel with its capacitance C_D. The amplifier has gain $-A$ and feedback capacitance C_f. The dashed resistor R_f is present in a continuous-discharge amplifier and absent in a reset amplifier.

The current through a capacitor is the rate of change of the charge on it, $i_C = C\,\mathrm{d}v_C/\mathrm{d}t$, where v_C is the voltage across its plates. The voltage across C_D is v_i; the voltage across C_f is $v_i - v_o$. Hence

$$i(t) = C_D\,\frac{\mathrm{d}v_i}{\mathrm{d}t} + C_f\,\frac{\mathrm{d}(v_i - v_o)}{\mathrm{d}t}. \tag{2}$$

In the Laplace domain[1] Eq. (2) becomes

$$I(s) = s\,C_D\,V_i(s) + s\,C_f\,[V_i(s) - V_o(s)], \tag{3}$$

where s is the Laplace variable.[2] The voltage across the feedback capacitor is the difference between its two plates. One plate is the input node, at v_i. The other is the output, and the amplifier inverts: $v_o = -Av_i$. When the input rises by v_i the output falls by Av_i, so the two plates move in opposite directions and the capacitor sees the sum of the two excursions,

$$v_i - v_o = v_i + A\,v_i = (1 + A)\,v_i. \tag{4}$$

The capacitor therefore draws $1 + A$ times the current that a grounded capacitor of the same value would draw at the same input voltage[3]. Substituting Eq. (4) in Eq. (3), and writing the transforms

[1]The Laplace transform of a function of time $x(t)$ is

$$X(s) = \int_0^\infty x(t)\,e^{-st}\,\mathrm{d}t,$$

where s is a complex variable. The transform of a derivative follows by integrating by parts:

$$\int_0^\infty \frac{\mathrm{d}x}{\mathrm{d}t}\,e^{-st}\,\mathrm{d}t = \Big[x(t)\,e^{-st}\Big]_0^\infty + s\int_0^\infty x(t)\,e^{-st}\,\mathrm{d}t = s\,X(s) - x(0).$$

The boundary term at infinity vanishes for any signal that grows slower than e^{st}. The term $x(0)$ is the initial value; for the response to a signal that starts from rest it is zero, and for noise it only shifts the mean. With that convention a derivative in time is a multiplication by s. Each term of Eq. (2) transforms separately, since the transform is linear and the capacitances are constants: $C_D\,\mathrm{d}v_i/\mathrm{d}t \to sC_DV_i(s)$, $C_f\,\mathrm{d}(v_i - v_o)/\mathrm{d}t \to sC_f[V_i(s) - V_o(s)]$ and $i(t) \to I(s)$.

[2]For a signal that is a steady sinusoid of angular frequency ω, every voltage and current can be written as a complex amplitude times $e^{j\omega t}$, for example $v_i(t) = \mathrm{Re}\,[V_i\,e^{j\omega t}]$, with $e^{j\omega t} = \cos\omega t + j\sin\omega t$. Then $\mathrm{d}(V_i e^{j\omega t})/\mathrm{d}t = j\omega\,V_i\,e^{j\omega t}$: differentiation multiplies the amplitude by $j\omega$, and the common factor $e^{j\omega t}$ cancels from every term of the equation. The Laplace variable restricted to the imaginary axis, $s = j\omega$, therefore describes the response to sinusoids, which is the frequency response. A general signal is a superposition of sinusoids, so the frequency response determines the response to anything. The magnitude $|V_i|$ is the amplitude of the sinusoid and the argument of V_i its phase.

[3]This is the Miller effect: seen from the input node, C_f behaves as a capacitance $(1 + A)C_f$ to ground.

in lowercase from here on,[4]

$$i = sC_D\, v_i + sC_f\,(v_i - v_o) = s\, v_i\,[C_D + (1+A)C_f]. \tag{5}$$

Equation (5) gives the input voltage produced by the pad current, $v_i = i/\{s[C_D + (1+A)C_f]\}$. The output is $-A$ times that:

$$v_o = -A\, v_i = -\frac{A}{s\,[C_D + (1+A)C_f]}\; i. \tag{6}$$

For a large gain, $AC_f \gg C_D$ and $A \gg 1$, the bracket is AC_f and the gain cancels:

$$v_o \xrightarrow{AC_f \gg C_D} -\frac{1}{sC_f}\; i. \tag{7}$$

Division by s is integration in time. For a large gain the output is

$$\boxed{v_o(t) = -\frac{1}{C_f}\int_0^t i(t')\,\mathrm{d}t' = -\frac{Q(t)}{C_f}} \tag{8}$$

the charge collected so far divided by the feedback capacitance.

Three properties follow from Eqs. (5) to (7).

- The input node is a virtual ground: $v_i = -v_o/A$ is small, so the pad stays at a fixed potential while charge flows into the amplifier.
- The output does not depend on C_D as long as $AC_f \gg C_D$. The gain is set by one capacitor —thus the name, charge sensitive amplifier—.
- Part of the charge stays on the pad. The input node is connected to two capacitances charged to the same voltage v_i: C_D to ground and, through the Miller effect, $(1+A)C_f$ to the output. Each takes charge in proportion to its capacitance,

  $$Q_D = C_D\, v_i, \qquad Q_f = (1+A)\, C_f\, v_i, \qquad Q = Q_D + Q_f, \tag{9}$$

  and only Q_f reaches the output. The fraction left on the pad is

  $$\frac{Q_D}{Q} = \frac{C_D}{C_D + (1+A)C_f}. \tag{10}$$

  With $C_f = 3\,\mathrm{fF}$ and $A = 1000$ the Miller capacitance is $3.0\,\mathrm{pF}$, comparable to a $1.25\,\mathrm{pF}$ pad, and 29 % of the charge never reaches the output; for $A = 10^4$ the fraction is 4 %. Either the open-loop gain is made large enough for this fraction to be negligible, or the calibration measures the actual output per ion and absorbs it.

The feedback resistor R_f of Fig. 1 lets the collected charge leak away with the time constant $\tau_f = R_f C_f$, so that the output returns to its resting value and the amplifier never saturates. The present NAUSICAA amplifier uses a transistor for this purpose, with a measured small-signal decay time of 19.5 s. This note proposes the alternative: remove the resistor and reset the amplifier with a switch, using the two-capacitor architecture of Section 7.

3 Sources of noise and their equivalent charge

The amplifier A of Fig. 1 is a transistor circuit. Fig. 6 shows a possible implementation (a cascode). The noise of the circuit is dominated by the input transistor (M1 in the figure)[5].

[4] In Eq. (3), $i(t)$ is the pad current and $I(s)$ its transform. Circuit analysis usually drops the uppercase and uses one symbol for both, because the equations are almost always written in the transformed domain. The rest of this note follows that practice: a symbol written with the argument t, such as $i(t)$ or $v_o(t)$, is a function of time; the same symbol written without argument or together with s, as in $i = sC_D v_i$, is the transform.

[5] The pad is directly connected to M1 gate and the noise of all the other transistors referred to that gate is suppressed by their gain.

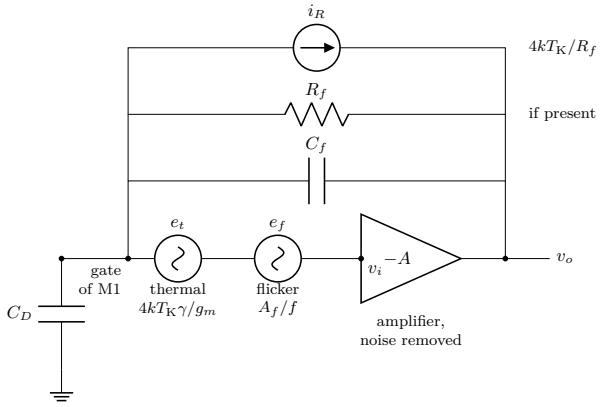


Figure 2: Noise model. The input transistor contributes two voltage sources referred to its gate, thermal e_t and flicker e_f; they are independent, so their mean squares add, $e_n^2 = e_t^2 + e_f^2$. They sit between the gate of M1, where the pad and the feedback branch meet, and the rest of the amplifier, which has no noise left once the sources are placed in front of it and whose input v_i is the virtual ground. A feedback resistor, if present, injects its current noise i_R directly into the gate node, in parallel with C_f.

The noise of M1 is represented by voltage sources in series with its gate, chosen to produce the same drain-current fluctuation as the transistor itself (the companion note on electronic noise [5] constructs them for the thermal part). What remains behind the sources is the amplifier with its noise removed, whose input is still the virtual ground. The gate of M1 is thus split in two: the physical gate node, where the pad and the feedback branch connect, and the input of the noiseless amplifier, with the noise sources between them.

Fig. 2 shows the resulting model with the relevant sources of noise. The companion note [5] derives their spectral densities from the thermal motion of charge.

- Thermal noise of the input transistor, $e_t^2 = 4kT_K\gamma/g_m$ with γ between 2/3 and about 1, and g_m the transconductance.
- Flicker noise of the same transistor, $e_f^2 = A_f/f$, from charge trapping at the oxide interface. It dominates below the corner frequency f_c, which for small transistors lies between kHz and MHz.
- Thermal noise of the feedback resistor, $i_R^2 = 4kT_K/R_f$. It is a current injected directly into the input node.

3.1 Voltage noise becomes a charge through the pad capacitance

The two transistor sources act at the same place and are treated together as one voltage source e_n. It sits between the gate of M1 and the noiseless amplifier behind it (Fig. 2). The gate is the summing node of the circuit: the pad capacitance and the feedback capacitor both connect to it. The question is how far the output moves when e_n takes a value, and which signal charge would move it by the same amount. Four steps answer it.

- The amplifier forces its own input, the terminal of e_n on its side, to the virtual ground. The gate, on the other terminal, is therefore not at zero but at e_n. The virtual ground follows the noise source.
- The pad capacitance has one plate on the gate and the other at ground, so it now holds the charge $C_D e_n$. Before the fluctuation it held none.

- That charge must come from somewhere. The gate node is isolated: the transistor draws no current and there is no resistor. The only element connected to it, besides the pad, is C_f. The charge $C_D e_n$ is therefore taken from the plate of C_f on the gate, which is left with $-C_D e_n$. Since that plate has e_n on its own side and v_o on the other, its charge is also $C_f(e_n - v_o)$. Conservation of charge on the gate node reads

$$C_D\, e_n + C_f\,(e_n - v_o) = 0.$$

- Solving for the output,

$$v_o = \frac{C_D + C_f}{C_f}\, e_n.$$

  The noise voltage appears at the output multiplied by $(C_D + C_f)/C_f$, the noise gain of the stage. For the NAUSICAA values, 1.25 pF and 3 fF, that gain is about 400.

A signal charge Q arriving at the pad moves the output by Q/C_f in magnitude, Eq. (8). The output excursion caused by e_n is therefore indistinguishable from a signal charge[6]

$$q_n = (C_D + C_f)\, e_n \simeq C_D\, e_n, \qquad \boxed{\; q_n^2(f) = C_D^2\, e_n^2(f) \;}. \tag{11}$$

The relation holds frequency by frequency, which is why it is written for the spectral densities: the noise charge spectrum is the noise voltage spectrum scaled by C_D^2. The same result follows from a source transformation: the voltage source e_n in series with the amplifier input, whose impedance is $Z_i = 1/(AC_f s)$, is equivalent to a current source $i_{An} = e_n/Z_i$ in parallel with it, and the charge that this current deposits on C_D is again $C_D e_n$. A feedback resistor does not alter this conversion as long as R_f is much larger than $|Z_i|$ at the frequencies of interest, so that it diverts no current.

Equation (11) is the central fact of low-noise front ends: the voltage noise of the transistor is multiplied by the capacitance seen at its gate. A pad of 5 pF gives four times the noise charge of a pad of 1.25 pF for the same transistor. The reference point of every noise figure in this note is the sensor: all noise is expressed as charge at the TopMetal pad, and that is what ENC means below.

3.2 Resistor noise becomes a charge by integration

The current i_R enters the input node and is integrated by C_f like a signal. In the frequency domain the integrated charge is $i_R/(j\omega)$, so

$$q_R^2(f) = \frac{4kT_\mathrm{K}}{R_f}\,\frac{1}{(2\pi f)^2}. \tag{12}$$

This spectrum rises without bound at low frequency. Integrated between a low limit f_l and a high limit f_h it gives $q_R^2 = \dfrac{4kT_\mathrm{K}}{4\pi^2 R_f}\left(\dfrac{1}{f_l} - \dfrac{1}{f_h}\right)$. The low limit is set by the measurement: a charge measured over a window T cannot see fluctuations slower than about $1/T$. Section 4 makes this precise.

3.3 Definition of ENC

A measurement is a linear operation on the output, with a transfer function $W(f)$ normalised so that a signal charge Q gives Q. The noise charge of the measurement is then

$$\boxed{\; \mathrm{ENC}^2 = \frac{1}{e^2}\int_0^\infty \left[C_D^2 e_n^2(f) + q_R^2(f)\right]\,|W(f)|^2\,\mathrm{d}f \;}. \tag{13}$$

Everything that follows is the choice of W.

[6] Had e_n been placed outside the loop, in series with the pad alone, the same steps would give $q_n = C_D e_n$ exactly: the feedback capacitor would then be at the virtual ground and hold no noise charge. The gate-referred noise of a transistor is inside the loop, hence the factor $C_D + C_f$. The difference is C_f, negligible against C_D here.

4 Integrating over a frame: the gated integrator

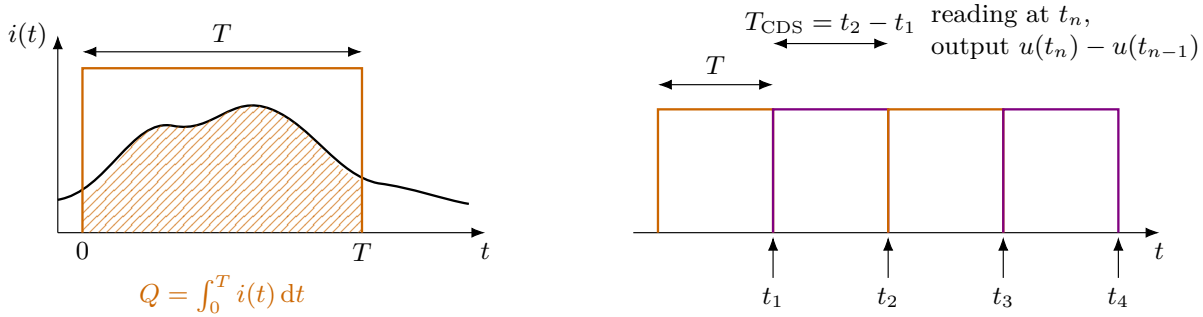


Figure 3: Left: the gated integrator. The current of one frame is integrated over a window of length T; the shaded area is the charge Q. Right: the frame sequence. Consecutive windows follow each other without gaps, the output is read at the end of each one, and the measurement is the difference of consecutive readings, T_{CDS} apart.

The ions of one frame arrive during a window of length T (Fig. 3, left). The amplifier integrates the current over that window:

$$Q = \int_0^T i(t)\,\mathrm{d}t. \tag{14}$$

As a filter acting on the current, this is a convolution with a rectangle of height 1 and length T. Its Fourier transform is

$$H(f) = T\,\mathrm{sinc}(\pi f T)\,e^{-j\pi f T}, \qquad \mathrm{sinc}\,x = \frac{\sin x}{x}, \tag{15}$$

equivalently $T\,\mathrm{sinc}(\omega T/2)$ with $\omega = 2\pi f$. The factor T is the DC gain of the window: a constant current i gives the charge iT. Applied to the noise charge, the same window must be normalised by $1/T$, because a constant charge on the input gives that charge and not T times it. The normalised weighting is therefore

$$|W(f)|^2 = \mathrm{sinc}^2(\pi f T). \tag{16}$$

Keeping the factor T in both the signal and the noise expression gives the same ENC, since it cancels in the ratio. The sinc function equals 1 at $f = 0$, has its first zero at $f = 1/T$ and decays as $1/f^2$ beyond. Two consequences follow.

- White noise above $1/T$ is averaged away. With $\int_0^\infty \mathrm{sinc}^2(\pi f T)\,\mathrm{d}f = 1/(2T)$, the thermal contribution is

  $$\mathrm{ENC}_t^2 = \frac{C_D^2\,e_t^2}{2T\,e^2}. \tag{17}$$

  A longer frame gives less thermal noise. For $C_D = 1.25\,\mathrm{pF}$, $e_t = 5\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$ and $T = 40\,\mathrm{ms}$ this is 0.14 electrons: the white part is irrelevant at frame time scales.

- Flicker noise below $1/T$ is not averaged (Fig. 4, left). The flicker contribution is $C_D^2 A_f \int \mathrm{sinc}^2(\pi f T)\,\mathrm{d}f/f$, which diverges logarithmically at low frequency. It must be cut at the lowest frequency the measurement can resolve, f_l, which is the inverse of the time between the reference level and the measurement. The result is of order $C_D^2 A_f\,\ln[1/(\pi f_l T)]$. The longer the amplifier runs between references, the larger this term.

The resistor term with the same window is dominated by its low-frequency end, where the window equals 1 and the spectrum of Eq. (12) rises as $1/f^2$. The cleaner statement is in the time domain: a white current noise integrated over T gives a charge variance

$$\boxed{\mathrm{ENC}_R^2 = \frac{4kT_{\mathrm{K}}}{R_f}\,\frac{T}{2\,e^2}}. \tag{18}$$

This term grows with the frame length, the opposite of the thermal term. If the NAUSICAA feedback transistor behaved as a resistor at 300 K with the value implied by its 19.5 s decay and $C_f = 3\,\mathrm{fF}$, about 6.5 TΩ, a 40 ms frame would carry 44 electrons of resistor noise. The transistor operates in weak inversion and its noise is the SPICE spectrum reported in `csa_design.tex`, which is the number to compare with. The decisive point is architectural: an amplifier that is reset by a switch has no R_f and no such term at all.

5 Correlated double sampling

The flicker term of Section 4 depends on how long the amplifier runs since its reference. Correlated double sampling removes that dependence. The measurement is the difference of two samples of the output taken T_{CDS} apart (Fig. 3, right),

$$y(t) = x(t) - x(t - T_{\mathrm{CDS}}). \tag{19}$$

This is a linear filter with impulse response $h(t) = \delta(t) - \delta(t - T_{\mathrm{CDS}})$ and transfer function

$$h(f) = 1 - e^{-j2\pi f T_{\mathrm{CDS}}} = e^{-j\pi f T_{\mathrm{CDS}}}\, 2j\, \sin(\pi f T_{\mathrm{CDS}}), \qquad |h(f)|^2 = 4\sin^2(\pi f T_{\mathrm{CDS}}). \tag{20}$$

At low frequency $\sin x \simeq x$ and

$$|h(f)|^2 \simeq 4\pi^2 f^2 T_{\mathrm{CDS}}^2 = \omega^2 T_{\mathrm{CDS}}^2, \qquad f \ll 1/T_{\mathrm{CDS}}. \tag{21}$$

The filter multiplies the spectrum by f^2 at low frequency. Three things disappear.

- A constant offset of the amplifier, V_{OFF}, is the same in both samples and cancels exactly. It is the $f = 0$ component.
- Slow drifts, with periods longer than T_{CDS}, are strongly attenuated.
- Flicker noise, whose $1/f$ spectrum would diverge, is multiplied by f^2 and becomes integrable. The low-frequency cut f_l of Section 4 is no longer needed.

Noise above $1/T_{\mathrm{CDS}}$ has twice the mean square after the subtraction, since the two samples are then independent and the variance of their difference is the sum of their variances. This is the price of the subtraction.

In NAUSICAA the reconstruction already forms the difference of consecutive frame readings, $u(t_n) - u(t_{n-1})$, in the digital domain. That difference is this filter with $T_{\mathrm{CDS}} = T$.

6 Total noise with both filters

Applying the window of Eq. (16) and the subtraction of Eq. (20) in sequence, the weighting function is their product and

$$\mathrm{ENC}^2 = \frac{C_D^2}{e^2} \int_0^\infty \mathrm{sinc}^2(\pi f T)\; 4\sin^2(\pi f T_{\mathrm{CDS}}) \left[e_t^2 + \frac{A_f}{f} \right] \mathrm{d}f, \tag{22}$$

The integrand has three factors: the sinc falling from 1, the noise profile falling as $1/f$ then flat, and the subtraction rising as f^2 then oscillating. The product is small at both ends and peaks near $1/T$ (Fig. 4).

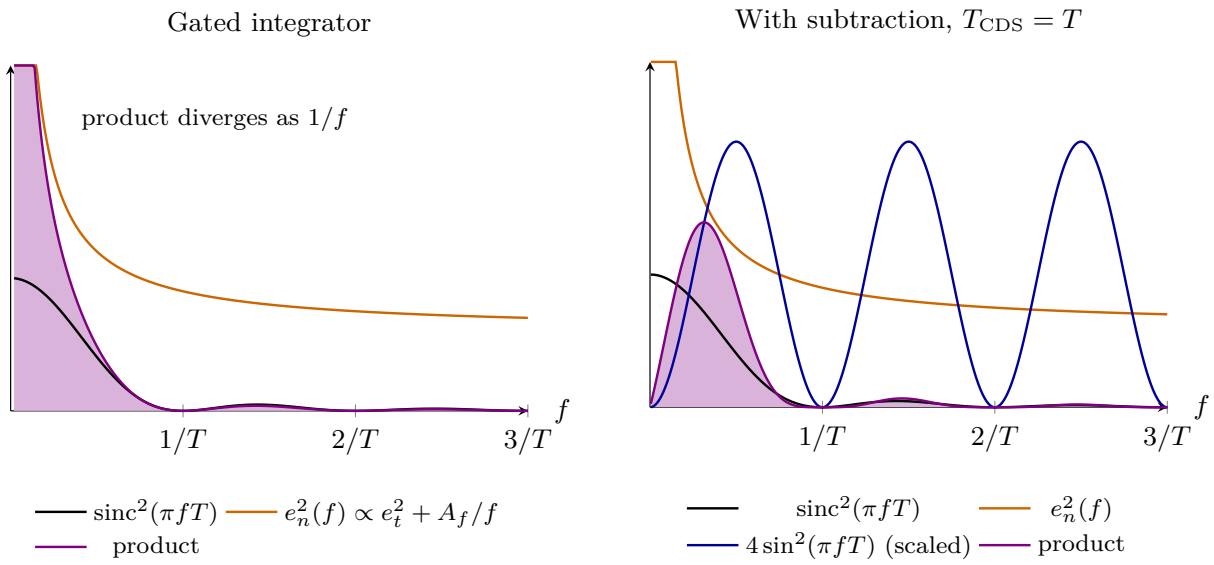


Figure 4: The three factors of Eq. (22) on a linear frequency axis, in arbitrary units, for a corner frequency $f_c = 0.5/T$. Left: the gated integrator alone. The window $\mathrm{sinc}^2(\pi fT)$ averages the white noise above $1/T$, but the product with the $1/f$ profile diverges at low frequency; its integral depends on the low-frequency cut. Right: with the subtraction of consecutive frames. The factor $4\sin^2(\pi fT)$ rises as f^2 and cancels the divergence; the product is now concentrated between 0 and $1/T$ and its integral is the closed form of Eq. (25).

For consecutive frames, $T_{\mathrm{CDS}} = T$, both integrals have closed forms. With $x = \pi fT$:

$$\int_0^\infty \mathrm{sinc}^2(\pi fT)\, 4\sin^2(\pi fT)\, \mathrm{d}f = \frac{4}{\pi T}\int_0^\infty \frac{\sin^4 x}{x^2}\,\mathrm{d}x = \frac{4}{\pi T}\cdot\frac{\pi}{4} = \frac{1}{T}, \tag{23}$$

$$\int_0^\infty \mathrm{sinc}^2(\pi fT)\, 4\sin^2(\pi fT)\, \frac{\mathrm{d}f}{f} = 4\int_0^\infty \frac{\sin^4 x}{x^3}\,\mathrm{d}x = 4\ln 2. \tag{24}$$

The two definite integrals are standard [6]. Hence

$$\boxed{\mathrm{ENC}^2 = \frac{C_D^2}{e^2}\left[\frac{e_t^2}{T} + 4\ln 2\, A_f\right]}. \tag{25}$$

The thermal term is twice that of a single window, Eq. (17), as expected for a difference of two independent averages. The flicker term is a pure number times $C_D^2 A_f$: it no longer depends on the frame length, nor on how long the amplifier has been running. With $f_c = 1\,\mathrm{kHz}$ and the values above it amounts to 2 electrons; the single-window estimate with a reference 100 s earlier would give 3. For a corner at 100 kHz the two numbers are 20 and 32. The subtraction removes the offset completely and reduces the flicker noise by the logarithm of the time since the last reference. In words, the largest noise component is strongly reduced; in numbers, the reduction is that logarithm.

The resistor term, if present, is of the same size with or without the subtraction: the difference of two integrals of a white current has the variance of Eq. (18). This is the reason the reset architecture is attractive at long frame times.

7 The two-capacitor architecture

A circuit can perform the subtraction of Section 5 without a second sample. Two equal capacitors $C_{f1} = C_{f2}$ alternate between the feedback position and a parking position at the reference voltage

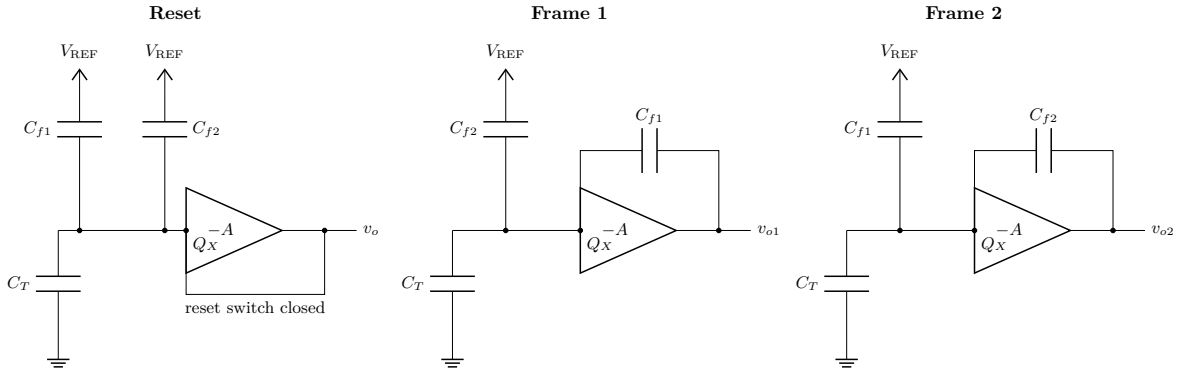


Figure 5: The two feedback capacitors through one cycle. C_T is the total capacitance at the input node, Q_X its charge. During reset both capacitors are connected between the input node and V_{REF} and the output is shorted to the input. In frame 1 C_{f1} is the feedback element while C_{f2} stays at V_{REF}; in frame 2 the roles are exchanged.

V_{REF} (Fig. 5). The amplifier has an offset V_{OFF}: it holds its input node at V_{OFF}, so the virtual ground sits at that voltage. The analysis is charge conservation at the input node, which is isolated except for the signal charge Q_i that arrives during each frame. The charge on C_T is $C_T V_{\mathrm{OFF}}$ in every phase and drops out of every balance below.

Reset. Both capacitors have one plate at the input node, at V_{OFF}, and the other at V_{REF}. The plate at the input node carries

$$Q_{X,R} = -(V_{\mathrm{REF}} - V_{\mathrm{OFF}})(C_{f1} + C_{f2}). \tag{26}$$

Frame 1. C_{f1} now connects the input node to the output v_{o1}; C_{f2} stays at V_{REF}. A charge Q_{i1} arrives. The input-node plates carry

$$Q_{X,S1} = -(V_{\mathrm{REF}} - V_{\mathrm{OFF}})\, C_{f2} + (V_{\mathrm{OFF}} - v_{o1})\, C_{f1}. \tag{27}$$

Conservation, $Q_{X,R} + Q_{i1} = Q_{X,S1}$, gives after cancelling the C_{f2} terms

$$-(V_{\mathrm{REF}} - V_{\mathrm{OFF}})C_{f1} + Q_{i1} = (V_{\mathrm{OFF}} - v_{o1})C_{f1} \quad \Longrightarrow \quad \boxed{v_{o1} = V_{\mathrm{REF}} - \frac{Q_{i1}}{C_{f1}}}. \tag{28}$$

The offset has cancelled: the two $V_{\mathrm{OFF}}C_{f1}$ terms are equal on both sides. The output is the reference voltage minus the collected charge divided by the feedback capacitance.

Frame 2. The capacitors exchange roles and Q_{i2} arrives:

$$Q_{X,S2} = -(V_{\mathrm{REF}} - V_{\mathrm{OFF}})\, C_{f1} + (V_{\mathrm{OFF}} - v_{o2})\, C_{f2}, \qquad Q_{X,S1} + Q_{i2} = Q_{X,S2}. \tag{29}$$

Substituting Eq. (28) for the C_{f1} term of $Q_{X,S1}$ and cancelling,

$$\boxed{v_{o2} = V_{\mathrm{REF}} - \frac{Q_{i1} + Q_{i2}}{C_{f2}}}. \tag{30}$$

Again every V_{OFF} term cancels. Three facts follow from Eqs. (28) and (30).

- Each output is referred to V_{REF}, a clean external voltage, and the amplifier offset never appears. This is the subtraction of Section 5 done by the capacitor swap: the parked capacitor carries the reference into the feedback position.

- The charge accumulates: v_{o2} holds $Q_{i1} + Q_{i2}$, and after n frames the output holds the sum of all charge since reset. The charge of one frame is the difference of consecutive outputs, as in NAUSICAA now, and the total charge of a track is read directly.
- The reset switch deposits a random charge of variance $kT_{\mathrm{K}}C$ on the capacitors. Because every later output is measured against the same reset, this charge cancels in differences, which is the original purpose of correlated double sampling in CCDs [4].

8 What follows for the design

Four design conclusions follow from the two-capacitor architecture. Each has a reason and a cost.

- **Offset and flicker cancellation in the analogue domain.** Eqs. (28) and (30) give the subtraction without storing and subtracting digital samples. The cost is the switch network at the input node, whose charge injection and leakage must be controlled.
- **Fewer ADC conversions.** One conversion per frame suffices, since the output already holds the accumulated charge. The present NAUSICAA plan samples every 100 µs and averages 50 samples over 5 ms before differencing; that average is the sinc window of Section 4 with $T = 5$ ms applied inside the frame.
- **A larger feedback capacitance.** Since the charge of a track accumulates, the full-scale output must hold the whole track and the converter must resolve one frame's charge within it. A larger C_f lowers the voltage per ion and so the range needed; it also lowers the closed-loop bandwidth, which Section 10 shows to be favourable for stability. The cost is a smaller signal per ion at the ADC.
- **A sigma-delta converter of 16 bits.** With accumulation the converter needs range and resolution together: 12 bits are 4096 levels, 16 bits are 65 536. A sigma-delta converter averages over its conversion time, which is one more sinc window and costs nothing at frame time scales.

9 The amplifier: a cascode with degeneration

A single-stage amplifier suited to this front end is shown in Fig. 6. The input transistor M1 converts the input voltage into a current; the cascode M2 passes that current to a high-impedance output node; the PMOS pair M4, M5 is the load. A resistor R in the source of M1 is the new element.

Transconductance with degeneration. Let v_i be the gate voltage and i the drain current of M1. The source sits at iR, so the gate-source voltage is $v_i - iR$ and $i = g_{m1}(v_i - iR)$. Solving,

$$i = \frac{g_{m1}}{1 + g_{m1}R}\, v_i \equiv G_m\, v_i. \tag{31}$$

For $g_{m1}R \gg 1$ the transconductance tends to $1/R$: it is then set by a resistor and no longer by the transistor's operating point, so the stage stays linear over a wide range of input. A large open-loop gain inside the feedback loop linearises the closed-loop response in the same way.

Gain. The current $G_m v_i$ flows into the output node, whose impedance is the parallel combination of the two cascodes. Looking up into M4, M5 the impedance is about $g_{m4}r_{d4}r_{d5}$; looking down into M2 it is about $g_{m2}r_{d2}$ times the impedance at the source of M2, which is the parallel of r_{d3} and the

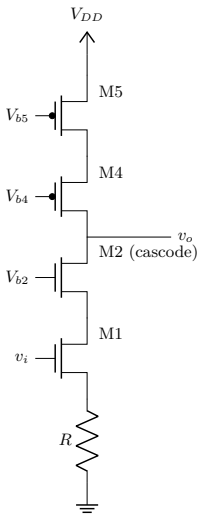


Figure 6: Single-stage amplifier: input transistor M1 with a degeneration resistor R in its source, cascode M2, and a cascoded PMOS load M4, M5. The output is the drain node of M2.

degenerated output of M1, $g_{m1}r_{d1}R$. Here r_d is the drain-source resistance of each transistor. The open-loop gain is

$$A = G_m\, Z_o, \qquad Z_o = (g_{m4}r_{d4}r_{d5}) \parallel \left[g_{m2}r_{d2}\left(r_{d3} \parallel g_{m1}r_{d1}R\right)\right]. \tag{32}$$

Each cascode multiplies an output resistance by a factor $g_m r_d$, which is why a single stage can reach gains of several thousand.

Noise. The resistor adds nothing to the transistor's own noise. It adds its own: the thermal voltage noise $4kT_{\mathrm{K}}R$ of R is in series with the source of M1, hence in series with the gate-source voltage, and appears at the input with that full value. Since $e_t^2 = 4kT_{\mathrm{K}}\gamma/g_{m1}$ for the transistor, the resistor adds the same noise as the transistor once $R \simeq \gamma/g_{m1}$. Degeneration is therefore likely to increase the noise budget; in current form the added term is $i_n^2 = 4kT_{\mathrm{K}}/R$.

10 The real amplifier: bandwidth and stability

A real amplifier has finite bandwidth. The simplest model replaces the ideal gain by a single-pole response

$$A(s) = \frac{A}{1 + s/\omega_0}, \tag{33}$$

where ω_0 is the dominant pole of the amplifier. For the stage of Fig. 6 that pole is formed by the output impedance and the capacitance at the output node: $\omega_0 = 1/(Z_o C_o)$, with C_o the sum of the load capacitance and of C_f as seen from the output. Inserting $A(s)$ in Eq. (6) with $C_D \ll AC_f$, and writing $Z_f = 1/(sC_f)$ for the impedance of the feedback capacitor, the ratio of output voltage to input current is

$$A_{CL}(s) = \frac{Z_f\, A(s)}{1 + A(s)} = Z_f\, \frac{A}{1 + A}\, \frac{1}{1 + \dfrac{s}{(1 + A)\,\omega_0}}. \tag{34}$$

The closed-loop response has a pole at $(1 + A)\omega_0$. Feedback extends the bandwidth by the loop gain. The product

$$\mathrm{GBP} = A\,\omega_0 = G_m Z_o \cdot \frac{1}{Z_o C_o} = \frac{G_m}{C_o} \;\simeq\; \frac{g_{m1}}{C_f} \tag{35}$$

is the gain-bandwidth product and equals the closed-loop bandwidth (Fig. 7). The last form holds when C_f dominates C_o and R is absent. With the pad capacitance included the exact closed-loop

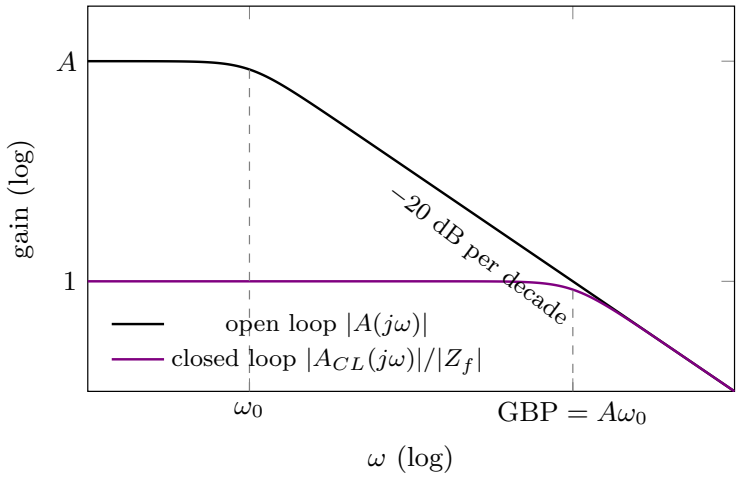


Figure 7: Open-loop gain of the single-pole amplifier and closed-loop gain of the charge amplifier, normalised to $|Z_f|$, on logarithmic axes ($A = 10^4$). The open-loop gain is flat up to ω_0 and falls by a factor 10 per decade beyond it, crossing unity at the gain-bandwidth product $A\omega_0$. Feedback trades the gain A for bandwidth: the closed-loop response is flat up to $(1 + A)\omega_0$ and then follows the open-loop curve.

pole is $g_{m1}/(C_D + C_f)$: the amplifier responds in a time $(C_D + C_f)/g_{m1}$, nanoseconds for picofarads and millisiemens.

A design constraint follows from Eq. (35). Low thermal noise requires a large g_{m1}; a high charge gain requires a small C_f; the two together give a very large GBP. With $g_{m1} = 1\,\text{mS}$ and $C_f = 3\,\text{fF}$ the ratio is $3 \times 10^{11}\,\text{rad/s}$. No single-stage amplifier is that fast, so the second pole of the stage, at the cascode nodes, falls inside the loop bandwidth and the phase margin is lost. Three remedies exist.

- The degeneration resistor lowers G_m and with it the GBP, at the noise cost of Section 9.
- A load capacitor C_L at the output lowers ω_0 and sets the dominant pole deliberately.
- A second voltage-gain stage after the integrator lets C_f be larger while the overall sensitivity per ion is preserved; the larger C_f lowers the GBP.

Each remedy trades bandwidth, which is not needed at frame time scales, against noise or complexity.

11 Pad capacitance and input impedance

Does a pad of 1 pF to 5 pF, DC coupled to the amplifier, deliver its charge to the amplifier? The answer is the input impedance of the real amplifier. From Eq. (5) the current drawn by the feedback branch is $sC_f(1 + A(s))v_i$, so

$$Z_{\text{in}}(s) = \frac{1}{sC_f\,[1 + A(s)]} = \frac{1 + s/\omega_0}{sC_f\,[1 + A + s/\omega_0]}. \tag{36}$$

Three regimes follow (Fig. 8).

- Below ω_0: $Z_{\text{in}} \simeq 1/[sC_f(1 + A)]$, a capacitor of value $(1 + A)C_f$: the Miller capacitance of Section 2.

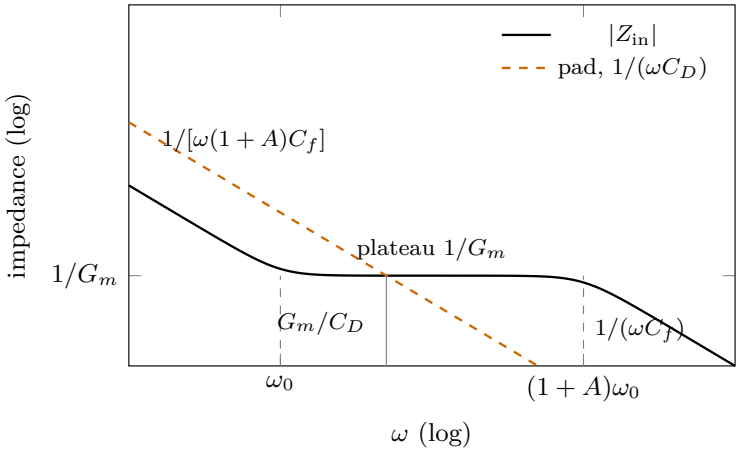


Figure 8: Magnitude of the input impedance of Eq. (36) on logarithmic axes, for $A = 10^4$ and $C_D = 400\,C_f$. Below ω_0 it is the Miller capacitance $(1 + A)C_f$; between ω_0 and $(1 + A)\omega_0$ it is the plateau $1/G_m$; above, the bare C_f. The dashed line is the impedance of the pad. The pad delivers its charge to the amplifier where its impedance lies above $|Z_{\text{in}}|$, that is below $\omega = G_m/C_D$.

- Between ω_0 and $(1 + A)\omega_0$: the numerator and denominator both grow with s and the magnitude is flat, $|Z_{\text{in}}| \simeq 1/(AC_f\omega_0) = 1/G_m$. The input impedance of a charge amplifier at these frequencies is the inverse transconductance of its input transistor.
- Above $(1 + A)\omega_0$: $Z_{\text{in}} \simeq 1/(sC_f)$, the bare feedback capacitor, since the amplifier no longer responds.

The pad delivers its charge to the amplifier when its own impedance $1/(\omega C_D)$ is much larger than Z_{in} at the frequencies of the signal. On the plateau the condition reads $\omega C_D \ll G_m$, that is, $\omega \ll G_m/C_D$. For $C_D = 5\,\text{pF}$ and $G_m = 1\,\text{mS}$ the limit is $2 \times 10^8\,\text{rad/s}$. Ion signals with 40 ms frames are seven orders of magnitude slower. The pad capacitance therefore does not limit charge transfer in NAUSICAA. Its effect is on the noise, through Eq. (11), and on the closed-loop bandwidth, through $g_{m1}/(C_D + C_f)$.

12 Relation to the present NAUSICAA readout

Table 1 sets the reset architecture of Section 7 beside the amplifier that this repository simulates and has calibrated.

Three points carry over regardless of which circuit is built.

- The frame difference already used in the NAUSICAA reconstruction is the filter of Eq. (20). The closed forms of Eq. (25) give the thermal and flicker noise of that reconstruction in terms of C_D, e_t, A_f and T. They are an independent check of the bucket noise covariance computed from the SPICE spectrum in `bucket_noise_covariance.jl`.
- The discharge-element term of Eq. (18) grows with the frame length and is untouched by the frame difference. Whether it matters for the present circuit is decided by the measured noise spectrum of the feedback transistor, which the SPICE model contains. The analytical expression says what to look for: a term proportional to T.
- The pad capacitance enters the noise as C_D^2 and the bandwidth as C_D/g_{m1}, and does not limit charge collection. The 1.25 pF of the calibrated circuit and the 1 pF to 5 pF range of the TopMetal pad lie in the same regime.

	Present NAUSICAA circuit	Reset architecture
Charge removal	Feedback transistor, decay 19.5 s, $C_f = 3$ fF	Reset switch; no discharge element
Reference per frame	Digital difference of consecutive readings; discharge term $\tau_f^{-1} \int u \, dt$ corrected in reconstruction	Analogue: capacitor swap, output referred to V_{REF}
Offset and flicker	Removed by the digital difference	Removed by the swap
Discharge-element noise	Present; measured in SPICE	Absent
Reset noise	None	$kT_{\mathrm{K}}C$, cancelled by the common reset
Nonlinearity	Feedback transistor; calibrated nonlinear model with 0.63 V guard	Linear integrator up to the output range; range set by C_f and the ADC
Conversion	12 bits every 100 µs, 50-sample average over 5 ms	One 16-bit sigma-delta conversion per frame
Charge per frame	From differences and the discharge correction	From differences; total charge read directly

Table 1: The two readout schemes side by side. The present values are those of `itaca_poo.tex` and `csa_design.tex`.

If the two-capacitor architecture is adopted, the itrack stage 2 of the readout description changes from feedback discharge and digital filtering to reset, swap and one conversion per frame, and the calibration must be repeated for the new circuit. The nonlinear model and its guard become unnecessary; the range limit becomes a reset. Section 13 reports a SPICE prototype of the reset architecture built on the present amplifier.

13 A SPICE prototype of the reset architecture

The architecture of Section 7 was simulated with the amplifier of the present NAUSICAA circuit, so that the two readouts can be compared with the same transistors, the same bias and the same device models. The prototype is called VH1; its decks are in `LTSpice/designs/VH1` and every number in this section is written by the scripts that run them.

13.1 The prototype

Figure 9 shows the circuit.

- The amplifier is M1 to M6 of the frozen `csa_v3` release with their sizes and biases: PMOS input M1 (220 µm/0.5 µm, 210 µA), NMOS cascode M4, load M3, current source M2 and a PMOS cascode pair M5, M6 at the output. The pad capacitance is 1.25 pF and the feedback capacitors are 3 fF, as in the present circuit. The discharge transistor Mf is removed.
- The capacitors park at $V_{\mathrm{REF}} = 0.886$ V, the output baseline of the present circuit, so that the output starts each cycle where it does today.
- The reset closes the loop through a level shift. In the present circuit the gate of M1 sits at 1.3 mV and the output at 0.886 V. A plain short of output to input would force the two to be equal, which this amplifier can only satisfy at 56 mV, outside its working range. The analysis of Section 7 needs the gate held at V_{OFF} with the loop closed and says nothing about the output value during reset, so any closed loop that holds the gate will do. A resistor of 8.85 MΩ carrying 0.1 µA provides the shift; the current must stay well below the 0.38 µA of the output stage. With it the output sits at 0.885 V and the gate at 0.3 mV during reset.

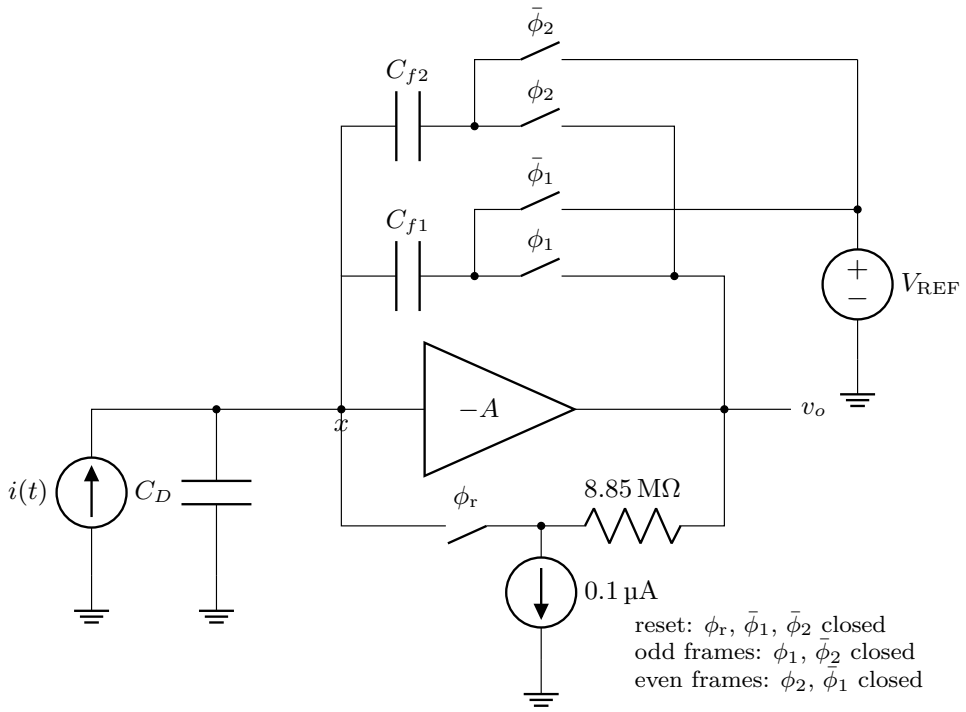


Figure 9: The VH1 prototype. The amplifier is the single stage of the present circuit (M1 to M6 of Fig. 6, without the resistor R); the discharge transistor is removed. The two feedback capacitors alternate between the output and V_{REF} through the switches $\phi_{1,2}$ and $\bar{\phi}_{1,2}$. The reset switch ϕ_{r} closes the loop through a level shift of 0.885 V, a resistor carrying 0.1 µA.

- The switches are ideal.[7]

13.2 Transfer function, charge sharing and range

The open-loop gain (Fig. 10, left) is $A_0 = 7605$ with a dominant pole at 30 kHz and unity gain at 94 MHz. The closed-loop bandwidth is 90 kHz with 3 fF and 1.1 MHz with 30 fF, both far above anything a 20 ms frame needs. The input impedance has the plateau of Section 11 at 689 Ω, that is $G_m = 1.45$ mS.

The charge sharing of Eq. (10) is larger than the pad alone would give. The input of the amplifier adds 4.34 pF to the 1.25 pF of the pad, 5.58 pF in all, against a Miller capacitance $(1 + A_0)C_f$ of 22.8 pF. The fraction of the charge that reaches the feedback capacitor is therefore 0.803, and the output moves by 42.9 µV per ion instead of $e/C_f = 53.4$ µV. The calibrated 40.2 µV per ion of the present circuit has the same origin. The added capacitance grows faster than linearly with the width of M1 and also with the width of the cascode M4, so it is a Miller term through the cascode node rather than the gate oxide of M1 alone.

The transients (Fig. 11) confirm the charge bookkeeping of Section 7.

- The output drops by 4.29 mV after frame 1 and 12.88 mV after frame 2, the same 42.9 µV per ion in both, and holds 12.88 mV through frame 3: the frame charge is the difference of two consecutive readings, Eq. (30).
- An offset of 50 mV inserted at the input changes none of these numbers, to the last digit

[7] Ideal switches have no charge injection, clock feedthrough or capacitance, and the reset noise $kT_{\mathrm{K}}C$ of Section 7 is not accessible to a small-signal noise analysis. The simulator's default leakage conductance of 1 pS across every open switch would discharge 3 fF in 3 ms; it is set to 10^{-18} S.

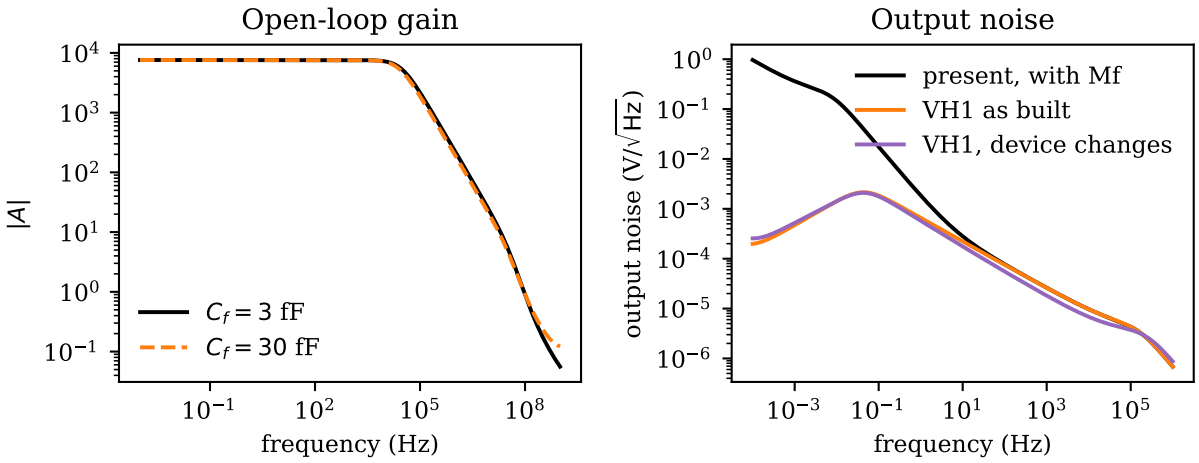


Figure 10: Left: open-loop gain of the amplifier, obtained in closed loop as $-v_o/v_i$. Right: output noise spectral density of the present circuit, of VH1 as built and of VH1 with the device changes of Section 13.4.

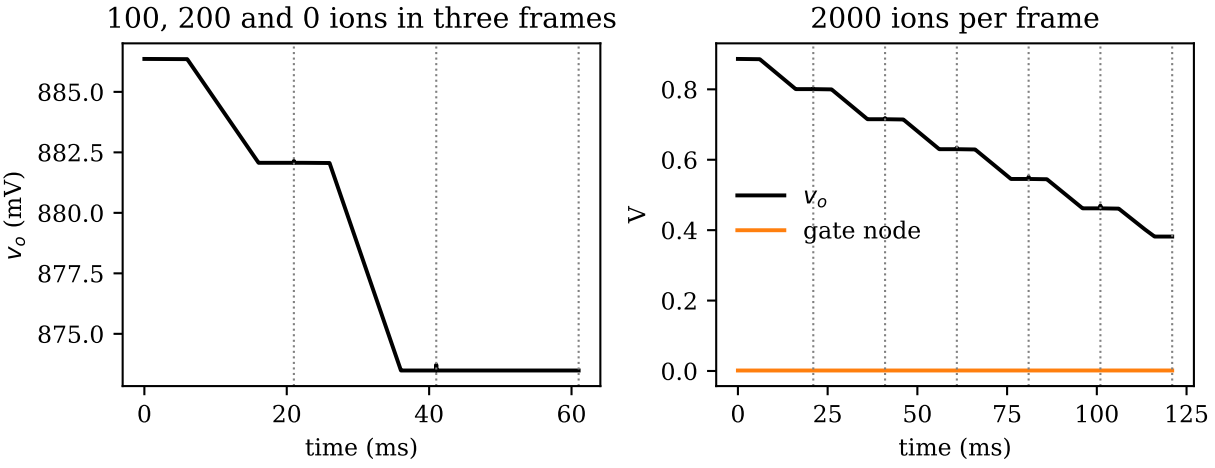


Figure 11: Left: reset at 1 ms, then 100 ions during frame 1, 200 during frame 2 and none during frame 3. Right: 2000 ions in each of six frames; the gate node stays at its bias while the output accumulates.

printed. With $V_{\mathrm{REF}} = 1.0\,\mathrm{V}$ the first frame is distorted: the working window of this output stage is about 0.4 V to 0.95 V.

- With 2000 ions per frame the output reaches 0.38 V after 12 000 ions, a drop of 0.50 V, with the gain per ion lower by 2 % at the end. There is no discharge, no guard and no nonlinear model to calibrate; the limit is the swing of the output stage.

13.3 Noise

The output noise spectra (Fig. 10, right) are identical above 10 Hz, where the transistors of the shared amplifier dominate. Below it the present circuit rises as $1/f^2$, to $1\,\mathrm{V}/\sqrt{\mathrm{Hz}}$ at 0.1 mHz: the current noise of the discharge transistor integrated by C_f, the resistor term of Section 4. In VH1 that term is absent and the spectrum flattens.

Table 2 gives the ENC obtained by applying the same weighting to both spectra, for 20 ms frames. Two checks anchor these numbers. The 5 ms weighting applied to the present circuit reproduces

Table 2: ENC in ions for 20 ms frames, from the simulated output noise spectra and the measured gain per ion of each circuit. The first row is the frame difference of samples taken at the frame boundaries; the second averages 5 ms around each boundary first, as the present reconstruction does; the third averages over the whole frame, the weighting of Eq. (22).

Weighting	Present circuit	VH1 as built	VH1, device changes
Frame difference of point samples	119.6	109.0	
Frame difference of 5 ms averages	50.2	40.2	28.3
Frame difference of 20 ms averages	36.7	27.5	19.7

the 50.2 ions of its calibration. And the difference between the first two columns matches the 26 % of the noise variance that the calibration attributed to the discharge transistor. Removing it is worth 10 ions at 20 ms; the remainder is the amplifier itself.

13.4 Device changes

The noise budget of the present circuit assigns 47 % of the variance to the current source M2 and 15 % to the load M3, against 6 % for the input transistor M1. Both inject their drain-current noise into the signal path, so their contribution scales with their transconductance. Table 3 reports single-device changes on VH1, each re-biased with the loop closed: the gate of M3 is tuned to keep the M1 current at 210 µA and the gate of M2 to keep the output at 0.886 V.

- **M2.** Lengthening the current source from 0.5 µm to 2 µm lowers its transconductance at the same current and takes the ENC from 40.2 to 34 ions, the largest single step. At 4 µm the gate bias approaches the supply and the open-loop gain falls; the benefit is lost.
- **M3.** The load helps only to 4 µm. A longer M3 needs more drain voltage than the 0.18 V available at the cascode node and leaves saturation, which collapses A_0. Raising the cascode bias would be the next step; it was not varied here.
- **M1.** Halving the input transistor to 110 µm cuts the capacitance it adds from 4.34 pF to 1.9 pF at equal ENC. At 55 µm the loss of transconductance costs more than the capacitance saves. The optimum is the usual one: gate capacitance near the pad capacitance.
- **M4.** A wider cascode does not reduce the input capacitance; it raises it, and with it the open-loop gain. Alone it is neutral. In combination it matters, because the longer M3 lowers

Table 3: Device changes on VH1. A_0 is the open-loop gain, C_{amp} the capacitance the amplifier adds at its input, and the fraction is the share of the charge that reaches the feedback capacitor. ENC in ions for 20 ms frames with 5 ms and full-frame averaging.

Variant	V_{nmos} (V)	V_{nload} (V)	A_0	C_{amp} (pF)	Fraction	5 ms	20 ms
VH1 as built	1.60	0.538	7605	4.3	0.80	40.2	27.5
M2 length 1 µm	1.61	0.538	6816	3.9	0.80	34.5	23.6
M2 length 2 µm	1.65	0.538	5720	3.3	0.79	34.0	23.3
M2 length 4 µm	1.73	0.538	4482	2.7	0.77	37.6	25.7
M3 length 4 µm	1.60	0.582	3669	2.4	0.75	38.4	26.2
M3 length 8 µm	1.60	0.661	1363	1.3	0.61	43.1	29.4
M3 length 16 µm	1.60	0.819	755	1.0	0.50	53.2	36.3
M4 width 5.8 µm	1.60	0.538	11016	6.0	0.82	39.7	27.1
M4 width 11.5 µm	1.60	0.538	14281	7.7	0.83	39.6	27.1
M4 width 23 µm	1.60	0.538	17213	9.1	0.83	39.6	27.0
M4 width 46 µm	1.60	0.538	19745	10.3	0.84	39.6	27.1
M1 width 110 µm	1.65	0.538	6353	1.9	0.86	39.4	27.1
M1 width 55 µm	1.73	0.538	4977	0.8	0.88	47.4	33.0
M2 2, M3 4, M1 110 µm	1.71	0.582	2379	0.9	0.77	30.3	21.0
M2 2, M3 4, M1 110, M4 23 µm	1.71	0.580	9269	2.6	0.88	28.3	19.7

A_0 and the charge fraction with it; the 23 µm cascode restores $A_0 = 9269$ and a fraction of 0.878.

The combination of the four changes gives 28.3 ions with 5 ms averaging and 19.7 with full-frame averaging, against 50.2 and 36.7 for the present circuit. The practical reset of Section 13.1 was verified on this combined circuit: 42.6 µV per ion from the frame difference.

13.5 Frame length

Table 4 evaluates the same spectra for frames of 10 ms, 20 ms, 40 ms and 80 ms. With full-frame

Table 4: ENC in ions against frame length, from the simulated spectra. Left: full-frame averaging before the frame difference. Right: 5 ms averages at the frame boundaries.

	Full-frame averaging				5 ms averages			
Circuit	10 ms	20 ms	40 ms	80 ms	10 ms	20 ms	40 ms	80 ms
Present circuit	33.8	36.7	42.5	52.4	41.6	50.2	60.4	74.1
VH1 as built	28.0	27.5	26.9	26.4	35.4	40.2	44.3	48.0
VH1 with device changes	19.6	19.7	19.8	20.0	24.6	28.3	31.7	34.8

averaging the ENC of VH1 does not depend on the frame length. This is Eq. (25): the thermal term falls as $1/T$ and is negligible already at 10 ms, the flicker term is a constant, and the resistor term that grows with T is absent. The present circuit has that term and gets worse with longer frames. With brief samples at the boundaries every circuit gets worse with T, because a wider spacing lets more low-frequency noise into the difference; the gain from long frames is conditional on integrating over them, which a sigma-delta converter does by construction.

The frame length is therefore free as far as the electronics are concerned, from 10 ms upwards. It is set by the tracking: the Z resolution of the ion track is about 1 mm, three times that is 3 mm, and 3 mm of drift is 20 ms. Longer frames buy signal per frame at the price of Z resolution; shorter ones buy Z resolution at the price of conversions and of a larger correction for the current pulse that straddles frame boundaries.

13.6 What remains open

- Switch charge injection, clock feedthrough and the reset noise are not in the simulation; they need transistor switches and a transient noise study.
- The cascode bias was kept at its present value; it limits the length of M3 and the full benefit of the device changes.
- The last row of Table 2 requires a converter that integrates over the frame. A converter that samples briefly at the swap realises the first row instead.
- A feedback capacitor of 30 fF would give ten times the range at the same ENC in ions, since signal and noise at the output scale together, and a cleaner phase margin; the price is converter resolution, Section 8.

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